Ontological Bedrock: Persistence, the Void, and the Fine-Structure Constant

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Abstract

We establish a minimal, defect-free foundation for determinate existence and apply it to two problems: the structural impossibility of absolute non-existence, and the derivation of the fine-structure constant.

Part I (Persistence Filter). Beginning from the single premise *something exists determinately*, we first give a precise four-condition definition of "determinately": not vague, not self-contradictory, not flux, and—crucially—not vacuous. A determinate identity predicate must be true of E and false of at least one thing; a predicate true of everything draws no distinction and carries no identity content. This fourth condition, absent from earlier treatments, closes the derivation of the first primitive without supplementary metaphysical assumption. From this definition we prove that exactly three structural primitives are forced: a state space (non-vacuity requires at least one alternative), a selective criterion (distinguishability is required by contrast), and an admissible transformation family (minimal self-identity is required by persistence). No smaller set suffices; no additional primitive is needed.

From these primitives we define a formal grounding map connecting selective criteria to base predicates, a *Globally Coupled Contradiction* (GCC) as a predicate inconsistent at its own grounding level, and two independent persistence conditions: Idempotent Closure (IC) and Energetic Viability (EV). Substrate capacity is defined combinatorially and independently of EV, eliminating the circularity present in earlier work. We prove: (1) a GCC makes the selective criterion undefined, eliminating determinate existence without *ex falso*; (2) IC and EV are logically independent; (3) the persistence constraint ψ^* satisfies its own conditions, with $\Delta(\psi^*) = \{\mathrm{id}\}$ as the unique formally sound choice; (4) IC failure implies either a GCC or a selectivity failure; (5) the falsification conditions are exhaustive. A Lean 4 machine-verification is publicly archived [6].

Part II (The Void). We apply the filter to its own limiting case. The Void—absolute non-existence with total potentiality and total absence simultaneously—contains a GCC (Theorem 15.3) and fails both filter conditions simultaneously (Theorem 15.4). Leibniz's question presupposes a coherent alternative that does not exist. The minimal resolution is a logical fixed point whose existence is guaranteed independently by Lawvere's fixed-point theorem and Kleene's second recursion theorem (Theorem 16.2).

Part III (Electromagnetic Coupling). In a consistent five-dimensional geometric framework, the EV condition applied to the electromagnetic vacuum forces a solvency equation whose unique solution is

$$\alpha = \frac{9}{16\pi^4}\left(\frac{\pi^5}{120}\right)^{1/4} \approx \frac{1}{137.036},$$

derived without assuming its value. The relative error against CODATA 2022 ($\alpha^{-1} = 137.035\,999\,177$) is 6×10^{-7}, at the threshold where QED loop corrections begin to matter.

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1 Introduction

1.1 Two Questions and a Common Answer

Why is there something rather than nothing? What does it take for anything to persist as a determinate entity? These questions have been treated as distinct. The first belongs to cosmology and theology; the second to metaphysics and the philosophy of science. This paper proves they have the same answer, derivable from the same structural constraint.

The constraint is simple: a form exists as a determinate, re-identifiable entity if and only if two conditions hold simultaneously. Its boundary must be stable under every transformation that constitutes it. The cost of maintaining that boundary must not exceed the sustaining capacity of its substrate. We call these Idempotent Closure (IC) and Energetic Viability (EV). Together they constitute the *persistence filter*.

The first question—Leibniz's question—is answered in Part II: the Void fails both filter conditions simultaneously and by structural necessity. It was never a coherent alternative to existence; the question contains a presupposition failure.

The second question—what it takes to persist—is answered in Part I: the filter is necessary and sufficient for determinate existence, self-applying, and exhaustively falsifiable.

Part III shows that the same filter, applied to the physical vacuum, fixes the electromagnetic coupling constant.

1.2 Structural Defects Repaired in This Version

Previous treatments of the persistence filter suffered from four structural defects.

Primitives not fully derived. The derivation of the state space previously relied on an unargued contrast requirement: a lone entity in a singleton domain would be "indistinguishable from nothing." That assertion was a logical gap; none of the three-condition definitions of "determinately" entailed it. The present paper repairs this by identifying a fourth necessary condition on determinate existence: non-vacuity (Section 3).

GCC elimination by *ex falso*. Earlier work invoked classical explosion. The present paper introduces the grounding map (Section 6) and proves elimination at the object level without any meta-logical step.

Circular substrate capacity. B_Σ was defined using the EV condition it was used to constrain. The present paper defines $B_\Sigma = 2^{|\Sigma|}$ from a finite reference support fixed independently of EV (Section 9).

Non-trivial self-application asserted without proof. Earlier work declared five named operations as admissible transformations without justification. The present paper takes $\Delta(\psi^*) = \{\mathrm{id}\}$ as the minimal formally sound choice and derives IC and EV from it directly (Section 11).

1.3 Relation to Prior Work

The logical structure draws on Spencer-Brown's mark and distinction [35], Von Foerster's eigenform theory [40], and Maturana–Varela's autopoiesis [27]. The cost formulation uses Landauer's principle [21] and the Margolus–Levitin bound [26]. The query-complexity formulation uses Kolmogorov–Chaitin complexity [20, 7] and decision-tree complexity [10]. The geometric coupling derives from Wyler's construction [41, 42]; Robertson's critique [34] is addressed directly in Section 17.10. The fixed-point results use Lawvere's theorem [22] and Kleene's second recursion theorem [19].

1.4 Scope Limitations

This paper does not derive physical constants other than α from first principles; they are empirical inputs. The law–mind corollary (Remark 16.6) does not imply panpsychism. The fixed-point proof does not presuppose rich structure prior to the first distinction; the required structure is supplied by the filter itself (Remark 16.3). The impossibility of the Void is structural, not epistemic (Remark 15.6).

2 Impossibility of Zero Distinction

Before characterising what it means to exist determinately, we establish that a domain in which no distinction is drawn is structurally impossible. This derivation takes a single input—Premise (P) below— and derives the necessity of distinction as a theorem, not a presupposition.

Premise (P). *Something exists determinately.*

Definition 2.1 (Zero-Distinction State)**.** *A* zero-distinction state *is a domain D such that for every predicate $P : D \to \{0, 1\}$,*

$$\forall\, x, y \in D : \; P(x) = P(y).$$

Equivalently, every predicate on D is either identically true or identically false.

Lemma 2.2 (Zero Distinction Implies Vacuity)**.** *In a zero-distinction state, every identity predicate is vacuous.*

Proof. Let $E \in D$. The identity predicate $I_E(x) := [x = E]$ must, by Definition 2.1, satisfy $I_E(x) = I_E(y)$ for all $x, y \in D$. Hence either $I_E(x) = 1$ for all x, or $I_E(x) = 0$ for all x. In the first case the predicate is universally true and distinguishes nothing. In the second case it fails to identify E at all. Both cases are vacuous. □

Lemma 2.3 (Vacuity Eliminates Determinate Existence)**.** *If all identity predicates on a domain are vacuous, no entity in that domain exists determinately.*

Proof. This follows immediately from Lemma 3.2 (Section 3): a vacuous identity predicate encodes zero information about the entity's specific identity, which is indeterminacy by contentlessness. □

Theorem 2.4 (Zero Distinction Is Structurally Impossible)**.** *A zero-distinction state cannot exist under Premise (P).*

Proof. A zero-distinction state entails that all identity predicates are vacuous (Lemma 2.2), which eliminates determinate existence (Lemma 2.3). This contradicts Premise (P). Therefore no zero-distinction state can obtain. □

Corollary 2.5 (Distinction Is Necessary)**.** *There exists at least one predicate P and elements x, y in the domain such that $P(x) \neq P(y)$.*

Definition 2.6 (First Distinction)**.** *A* first distinction *is a non-vacuous predicate whose existence is guaranteed by Corollary 2.5.*

Theorem 2.7 (Distinction Is Forced, Not Presupposed)**.** *The existence of at least one distinction follows necessarily from Premise (P). No external capacity to draw a boundary is assumed; its absence is structurally impossible.*

Proof. Theorem 2.4 and Corollary 2.5. □

Remark 2.8 (Closure of the Derivation). This section removes the only remaining external primitive. The persistence filter (Sections 3–14) operates on distinctions; Theorem 2.7 establishes that those distinctions exist necessarily from Premise (P) alone. Spencer-Brown's primary mark [35] and Aristotle's observation that a predicate applying to everything signifies nothing [2] are recovered here as formal consequences, not independent assumptions. The derivation chain is now:

$$\text{(P)} \implies \text{zero distinction impossible} \implies \text{distinction necessary} \implies \text{filter.}$$

3 What "Determinately" Means

Before deriving anything, we must state precisely what the premise asserts. The claim *something exists determinately* requires a precise characterisation of "determinately." We give it as four jointly necessary conditions.

Definition 3.1 (Determinate Existence — Preliminary)**.** *An entity E exists determinately only if all four conditions hold:*

(i) ***Not vague.*** *There is a fact of the matter about whether any predicate applies to E; no predicate applies to E only to a degree.*

(ii) ***Not self-contradictory.*** *No predicate both applies and fails to apply to E at the same time and in the same respect.*

(iii) ***Not flux.*** *E's identity does not drift indeterminately; it is stable under the transformations constitutive of its role.*

(iv) ***Not vacuous.*** *The identity predicate $I_E(x) := [x = E]$ is non-vacuously satisfied: $I_E(E) = 1$ and there exists at least one x in the domain with $I_E(x) = 0$.*

Conditions (i)–(iii) are familiar. Condition (iv) is new and requires justification.

Lemma 3.2 (Vacuity Entails Indeterminacy)**.** *If the identity predicate I_E is true of every element of its domain, then E does not exist determinately.*

Proof. Suppose $I_E(x) = 1$ for all x in the domain D. Then for any two candidates $x, y \in D$, $I_E(x) = I_E(y)$. The predicate I_E does not distinguish E from any other element of D; in particular, it does not distinguish E from any putative non-E. A predicate that applies to everything carries no information about what it picks out: it has no discriminative content.

Formally, if I_E is vacuously true, then the Kolmogorov complexity of I_E relative to any encoding of D is $O(1)$—the constant program "output 1"—regardless of what E is. The predicate therefore contains zero information about E's specific identity. An entity whose identity is encoded by a zero-information predicate has no determinate identity. This is a form of indeterminacy distinct from vagueness (i), self-contradiction (ii), and flux (iii): it is indeterminacy by contentlessness. □

Remark 3.3 (Aristotle's Observation Recovered). Aristotle observes in the *Metaphysics* [2] that a predicate applying to everything signifies nothing. Lemma 3.2 is the formal derivation of this observation from the concept of determinate identity, rather than its assertion as a first principle.

Remark 3.4 (Independence of the Four Conditions). The four conditions are mutually independent:

- An entity can be non-vacuous but vague: its identity predicate has borderline cases. (Excluded by (i), not by (iv).)
- An entity can be non-vacuous but self-contradictory: it satisfies both P and $\neg P$ for some predicate P. (Excluded by (ii), not by (iv).)
- An entity can be non-vacuous but drifting: its identity predicate returns different values under iterated re-application. (Excluded by (iii), not by (iv).)
- An entity can satisfy (i)–(iii) but be vacuous: a single entity in a singleton domain, not vague, not contradictory, not drifting, whose identity predicate is trivially universal. (Excluded by (iv), not by (i)–(iii).)

All four are therefore necessary; none is redundant.

Remark 3.5 (The Fourth Condition Is Not Stipulative). An objector might argue: adding condition (iv) amounts to building contrast into the premise, making the derivation of a two-element state space circular. This objection fails. Condition (iv) does not assert "there exists an alternative to E" as a bare ontological posit. It asserts that a determinate predicate—a piece of logical structure—must have non-trivial extension. This is an analytic claim about what predicates are: a predicate with no negative instances is not a predicate but a tautology. The existence of an alternative φ is then a consequence of the non-vacuity of I_E, not its presupposition.

4 Derivation of the Three Primitives

We take as our sole premise:

> **Premise (P).** Something exists determinately (in the sense of Definition 3.1).

4.1 First Primitive: The State Space

Proposition 4.1 (State Space is Forced)**.** *Determinate existence requires a non-empty set S with $|S| \geq 2$.*

Proof. Let E satisfy (P). By condition (iv) of Definition 3.1, $I_E(E) = 1$ and there exists at least one x with $I_E(x) = 0$. Call that element φ; then $\varphi \neq E$. Let S be the set containing E and all such elements. Then S is non-empty and $|S| \geq 2$. □

Remark 4.2. The earlier version of this argument asserted that a lone entity would be "indistinguishable from nothing." That assertion was a logical gap: the three-condition definition of "determinately" did not entail it. The present proof requires no such assertion. Condition (iv) directly forces the existence of φ; the argument is now deductive, not rhetorical.

4.2 Second Primitive: The Selective Criterion

Proposition 4.3 (Selective Criterion is Forced)**.** *Determinate existence requires a function $C_E : S \to \{0, 1\}$ with $C_E(E) = 1$ and $C_E(\varphi) = 0$ for at least one $\varphi \in S$.*

Proof. By Proposition 4.1, S contains E and at least one $\varphi \neq E$. For E to be *this* rather than φ, there must be a rule distinguishing E from φ. By condition (i) of Definition 3.1 (not vague), there is a fact of the matter about whether any predicate applies to E: the distinguishing rule is bivalent. It is therefore a function $C_E : S \to \{0, 1\}$ with $C_E(E) = 1$ and $C_E(\varphi) = 0$ for at least one φ. The non-universality condition follows from condition (iv): the identity predicate is already non-vacuous. □

4.3 Third Primitive: The Admissible Transformation Family

Proposition 4.4 (Transformation Family is Forced)**.** *Determinate existence requires a set $\Delta(E) \subseteq S^S$ satisfying: (i) $\Delta(E) \neq \emptyset$; (ii) $\mathrm{id}_S \in \Delta(E)$; (iii) $\Delta(E)$ is closed under composition.*

Proof. Determinate existence is not momentary. By condition (iii) of Definition 3.1 (not flux), E's identity is stable under the transformations constitutive of its role. The minimal such transformation is id_S, which trivially preserves $C_E(E) = 1$. Hence $\Delta(E)$ is non-empty and contains the identity. Closure under composition is the formal definition of what makes a collection of transformations constitute a single coherent role. A family not closed under composition would not define a stable role: applying two role-constitutive transformations in sequence could exit the role, making the role incoherent. The minimal admissible family satisfying this definition is $\{\mathrm{id}_S\}$. □

4.4 Necessity and Sufficiency

Theorem 4.5 (The Three Primitives are Jointly Necessary and Sufficient)**.** *Given Premise (P) and Definition 3.1, the three primitives are jointly necessary for determinate existence. No smaller set suffices. No additional primitive is required beyond what is entailed by formalising these three.*

Proof. Necessity. Propositions 4.1–4.4. Removing any one primitive destroys determinate existence.

Sufficiency. Given S, C_E, and $\Delta(E)$, a form is fully specified as a determinate entity. Any additional candidate primitive—bare particulars, haecceities, tropes—either reduces to one of these three or introduces structure not entailed by (P) and Definition 3.1, hence not a necessary primitive. □

5 Formal Definitions

Definition 5.1 (State Space)**.** *A state space is a non-empty set S with $|S| \geq 2$. No metric, topology, or algebraic structure is presupposed.*

Definition 5.2 (Form and Selective Criterion)**.** *A* form *is a pair (ψ, C_ψ) where $\psi \in S$ and $C_\psi : S \to \{0, 1\}$ is a selective criterion satisfying:*

(i) $C_\psi(\psi) = 1$,

(ii) $\exists\, \varphi \in S,\ C_\psi(\varphi) = 0$.

A criterion failing (ii) is universal; it draws no distinction. Condition (ii) is the formal content of Aristotle's observation [2] that determinate predication requires excluding at least one alternative.

Remark 5.3 (Bivalence Is Forced, Not Stipulated). The predicate structure $C_\psi : S \to \{0, 1\}$ might appear to be an arbitrary choice. We show it is the only structure consistent with Definition 3.1 and the finite grounding requirement of Section 6.

Any extension to a richer value set V with $|V| > 2$ falls into one of three cases.

Case 1: V contains an indeterminate value μ. A predicate P with $P(\psi) = \mu$ violates condition (i) of Definition 3.1 (not vague): there is no fact of the matter about whether P applies to ψ. Excluded.

Case 2: $V = \{0, 1, v\}$ where v is a third determinate value. The selective criterion requires $C_\psi(\psi) = 1$ (Definition 5.2(i)). If additionally $C_\psi(\psi) = v$ for $v \neq 1$, the grounding structure has a base predicate simultaneously assigning two distinct values to ψ, which is exactly a GCC (Definition 7.1). By Theorem 7.3, a GCC eliminates determinate existence. If v is instead a separate identity value that does not conflict with 1, it requires a new selective criterion to distinguish v-states from 1-states and 0-states. That criterion is itself a predicate over some value set, restarting the same question. The minimal resolution of this regress is bivalent predication: $\{0, 1\}$.

Case 3: Higher-order or type-theoretic predicates. A predicate whose codomain is itself a set of predicates either bottoms out in bivalent truth values at the base type (recovering $\{0, 1\}$), or generates an infinite regress of types with no base level. An infinite grounding set violates the finite grounding requirement (Section 6): evaluation cost becomes unbounded, eliminating EV.

In all three cases $\{0, 1\}$ is the unique value set consistent with Definition 3.1 and the filter. Bivalence is not selected; it is forced.

Definition 5.4 (Admissible Transformation Family)**.** *An admissible transformation family for ψ is a set $\Delta(\psi) \subseteq S^S$ satisfying: (i) $\Delta(\psi) \neq \emptyset$; (ii) $f, g \in \Delta(\psi) \Rightarrow f \circ g \in \Delta(\psi)$; (iii) $\mathrm{id}_S \in \Delta(\psi)$. $\Delta(\psi)$ is fixed prior to evaluation.*

Remark 5.5 (Why "Fixed Prior to Evaluation"). If $\Delta(\psi)$ could be chosen after evaluating which transformations preserve ψ, any form could be made to seem persistent by discarding disconfirming transformations. Fixing $\Delta(\psi)$ prior to evaluation parallels Tarski's requirement [38] that the truth predicate be defined relative to a fixed language before any object-level sentence is evaluated.

6 The Grounding Map

Definition 6.1 (Grounding Set and Grounding Function)**.** *A grounding set for C_ψ is a finite family $\mathcal{P} = \{P_1, \ldots, P_k\}$ of predicates $P_i : S \to \{0, 1\}$. C_ψ is* grounded *in $\mathcal{P}$ if there exists a Boolean function $F : \{0, 1\}^k \to \{0, 1\}$ such that for all $x \in S$:*

$$C_\psi(x) = F\Big(P_1(x), \ldots, P_k(x)\Big).$$

Finiteness of $\mathcal{P}$ is required: an infinite grounding set would make evaluation cost unbounded.

Definition 6.2 (Grounding Level)**.** *The* grounding level *of (ψ, C_ψ) with grounding structure $(F, \{P_1, \ldots, P_k\})$ is the tuple $(P_1(\psi), \ldots, P_k(\psi)) \in \{0, 1\}^k$.*

7 The Globally Coupled Contradiction

Definition 7.1 (Globally Coupled Contradiction)**.** *A form (ψ, C_ψ) with grounding structure $(F, \{P_1, \ldots, P_k\})$ contains a* Globally Coupled Contradiction *(*GCC*) if there exists $i \in \{1, \ldots, k\}$ such that $P_i(\psi) = 1$ and $P_i(\psi) = 0$.*

The qualifier *globally coupled* distinguishes this from contradictions that paraconsistent systems tolerate [32]: those systems survive precisely by providing a stable partition. A GCC is a contradiction for which no such partition exists.

Theorem 7.2 (GCC Makes $C_\psi(\psi)$ Undefined)**.** *If (ψ, C_ψ) contains a* GCC*, then $C_\psi(\psi)$ is undefined.*

Proof. Let i be the index with $P_i(\psi) = 1$ and $P_i(\psi) = 0$. The grounding level must be an element of $\{0, 1\}^k$ for F to be applicable. But $P_i(\psi)$ must simultaneously be 1 and 0; no element of $\{0, 1\}$ satisfies both. Hence $(P_1(\psi), \ldots, P_k(\psi)) \notin \{0, 1\}^k$. F cannot be applied. Therefore $C_\psi(\psi) = F(\ldots)$ is undefined. □

Theorem 7.3 (GCC Eliminates Determinate Existence)**.** *If (ψ, C_ψ) contains a* GCC*, then it does not exist determinately in any substrate.*

Proof. By Theorem 7.2, $C_\psi(\psi)$ is undefined. Definition 5.2(i) requires $C_\psi(\psi) = 1$. Undefined $\neq 1$. Condition (i) fails. □

Remark 7.4 (No *Ex Falso*). This proof uses only: (a) the grounding map makes $C_\psi(\psi)$ undefined; (b) a form requires $C_\psi(\psi) = 1$; (c) undefined $\neq 1$. No classical explosion or excluded middle is used. The result holds within paraconsistent settings [32].

8 Idempotent Closure

Definition 8.1 (Idempotent Closure)**.** *A form* (ψ, C_ψ) *satisfies* Idempotent Closure *(*IC*) with respect to* $\Delta(\psi)$ *if for every* $\delta \in \Delta(\psi)$ *and every* $n \in \mathbb{N}$*:*

$$C_\psi(\delta^n(\psi)) = 1,$$

where $\delta^0 = \mathrm{id}$ *and* $\delta^n = \delta \circ \delta^{n-1}$*.*

Remark 8.2. For $n = 0$: $\delta^0(\psi) = \psi$, so IC subsumes Definition 5.2(i). The $\forall n$ quantifier is essential: IC at depth 1 does not imply IC at depth 2. This is the formal analogue of Von Foerster's eigenform [40] and the recursive closure of Maturana–Varela's autopoiesis [27].

Proposition 8.3 (IC Failure Implies GCC or Selectivity Failure)**.** *If* (ψ, C_ψ) *fails* IC *with respect to* $\Delta(\psi)$*, then either: (a)* (ψ, C_ψ) *contains a* GCC*, or (b)* C_ψ *is not a selective criterion for* ψ*.*

Proof. IC fails means $\exists\, \delta \in \Delta(\psi)$, $n \in \mathbb{N}$ with $C_\psi(\delta^n(\psi)) \neq 1$.

Case $n = 0$: $C_\psi(\psi) = 0$, violating Definition 5.2(i). Selectivity failure (b).

Case $n \geq 1$: Let $\psi' = \delta^n(\psi)$. If C_ψ has a consistent grounding at ψ', then $C_\psi(\psi') = 0$ is well-defined: selectivity failure (b). If the grounding at ψ' has a base predicate inconsistent at ψ', then $C_\psi(\psi')$ is undefined by Theorem 7.2: GCC (a). □

9 Substrate Capacity and Energetic Viability

9.1 Substrate Capacity Defined Independently

Definition 9.1 (Finite Reference Support)**.** *A finite set* $\Sigma \subseteq S$ *is a* reference support *for* C_ψ *if for all* $x, y \in S$*:*

$$\Big(\forall\, s \in \Sigma,\ d(x, s) = d(y, s)\Big) \Rightarrow C_\psi(x) = C_\psi(y).$$

Definition 9.2 (Substrate Capacity)**.** *Let* Σ *be a reference support for* C_ψ*. The substrate capacity is* $B_\Sigma := 2^{|\Sigma|}$*.*

Remark 9.3 (No Circularity). B_Σ is a combinatorial quantity depending only on Σ and d. It is defined before EV and does not reference EV or the maintenance cost W. The definition chain is acyclic.

9.2 Query Trees, Finite Support, and the Computational Precision Result

Lemma 9.4 (Tree Computes Within Its Node Set)**.** *If* $p = \mathrm{eval}(T, \cdot)$*, then* $\mathrm{nodes}(T)$ *is a reference support for* p*.*

Proof. Structural induction on T. Base case trivial. Inductive step: at $\mathrm{Node}(s, T_L, T_R)$, $s \in \mathrm{nodes}(T)$ so $d(x, s) = d(y, s)$; both descend to the same subtree; by induction the same output is returned. □

Theorem 9.5 (Selective Criterion Has Finite Support)**.** *A predicate $p : S \to \{0, 1\}$ has a finite reference support if and only if it is computable by some query tree.*

Theorem 9.6 (GCC Forces Unbounded Query Cost)**.** *If (ψ, C_ψ) contains a* GCC*, then for every finite B_Σ and every candidate selective criterion C'_ψ, the query depth required to compute C'_ψ exceeds B_Σ.*

Remark 9.7 (Unification of the Two Conditions)*.* Theorem 9.6 shows that GCC directly forces unbounded query complexity, which directly violates EV. The path is: GCC $\Rightarrow Q(C_\psi) = \infty \Rightarrow W > B_\Sigma \Rightarrow$ not determinately existent.

9.3 Kolmogorov Complexity and Maintenance Cost

Definition 9.8 (Kolmogorov Complexity)**.** *Fix a universal prefix-free Turing machine U. $K(x)$ is the bit-length of the shortest program p with $U(p) = x$. Standard properties [25, 20, 7]: $(K1)$ Universality; $(K2)$ Subadditivity; $(K3)$ $K(n) = \Theta(\log n)$; $(K4)$ $K(\underbrace{x, \ldots, x}_{n}) \leq K(x) + c$.*

Definition 9.9 (Maintenance Cost)**.**

$$W(\psi, \delta, n) := K\Big(\psi,\ \delta(\psi),\ \ldots,\ \delta^n(\psi)\Big).$$

Definition 9.10 (Energetic Viability)**.** (ψ, C_ψ) *satisfies* Energetic Viability *(*EV*) with respect to $\Delta(\psi)$ and capacity B_Σ if*

$$\forall\, \delta \in \Delta(\psi),\ \forall\, n \in \mathbb{N} : \quad W(\psi, \delta, n) \leq B_\Sigma.$$

Definition 9.11 (Determinate Existence — Formal)**.** (ψ, C_ψ) exists determinately *if and only if: (i) C_ψ is a selective criterion for ψ (Definition 5.2); (ii) (ψ, C_ψ) satisfies* IC *(Definition 8.1); (iii) (ψ, C_ψ) satisfies* EV *(Definition 9.10). If any condition fails, the expression "ψ exists" lacks a determinate referent.*

Theorem 9.12 (Unbounded Cost Eliminates Determinate Existence)**.** *If $\exists\, \delta \in \Delta(\psi)$ with $W(\psi, \delta, n) \to \infty$, then (ψ, C_ψ) does not exist determinately for any finite B_Σ.*

Proof. For any finite B_Σ, $\exists\, n^*$ with $W(\psi, \delta, n^*) > B_\Sigma$. EV fails. □

10 Independence of IC and EV

Theorem 10.1 (Independence)**.** IC *and* EV *are logically independent: each is satisfiable while the other fails.*

Proof. **IC holds, EV fails.** Let $S = \mathbb{N}$, $\psi = 0$, $C_\psi(n) = 1$ iff $n = 0$. Define $\delta_0 : n \mapsto 0$: $\delta_0^n(\psi) = 0$, $C_\psi(0) = 1$ for all n. IC holds. Add δ_h encoding n steps of history: by $(K3)$, $W(\psi, \delta_h, n) = \Theta(\log n)$, unbounded. EV fails.

EV holds, IC fails. Let $S = \{0, 1\}$, $\psi = 0$, $C_\psi(x) = 1$ iff $x = 0$, $\delta : x \mapsto 1 - x$. Then $\delta(\psi) = 1$, $C_\psi(1) = 0$: IC fails. The alternating sequence has description length $O(1)$, so EV holds. □

11 Self-Application Without Regress

Definition 11.1 (ψ^* as a Finite Syntactic Object)**.**

$$\psi^* := (1, 1, 1, 1, 1, 1) \in \{0, 1\}^6,$$

*with components: has-state-space, has-form, has-transform-family, has-*IC*, has-*EV*, has-falsification. The selective criterion is* $C_{\psi^*}(x) := \min(x_1, \ldots, x_6)$.

Definition 11.2 ($\Delta(\psi^*)$)**.** $\Delta(\psi^*) := \{\mathrm{id}_{\{0,1\}^6}\}$.

Remark 11.3. Any non-trivial transformation changing a component from 1 to 0 violates IC by definition. The minimal admissible family is therefore $\{\mathrm{id}\}$.

Theorem 11.4 (ψ^* Satisfies IC)**.** (ψ^*, C_{ψ^*}) *satisfies* IC *with respect to* $\Delta(\psi^*)$.

Proof. $\Delta(\psi^*) = \{\mathrm{id}\}$. For $\delta = \mathrm{id}$ and any n: $\delta^n(\psi^*) = \psi^*$. $C_{\psi^*}(\psi^*) = 1$. □

Theorem 11.5 (ψ^* Satisfies EV)**.** (ψ^*, C_{ψ^*}) *satisfies* EV *with respect to* $\Delta(\psi^*)$ *and* $B^*_\Sigma = K(\psi^*) + c + 1$ *for some fixed* c.

Proof. The iteration sequence is $(\psi^*, \ldots, \psi^*)$ (n+1 times). By ($K4$): $W(\psi^*, \mathrm{id}, n) \leq K(\psi^*) + c$ for some c independent of n. Set $B^*_\Sigma = K(\psi^*) + c + 1$. Since $\psi^* \in \{0, 1\}^6$, $K(\psi^*) \leq 6 + O(1)$, so B^*_Σ is a small fixed integer. EV holds. □

Theorem 11.6 (ψ^* Exists Determinately)**.** (ψ^*, C_{ψ^*}) *exists determinately.*

Proof. (i) Selectivity: by definition. (ii) IC: previous theorem. (iii) EV: previous theorem. □

Remark 11.7 (Denial Presupposes the Constraint)*.* Any determinate denial of ψ^* must itself be a determinate assertion. It requires a state space, a selective criterion, and a transformation family—precisely the six components ψ^* records. A determinate denial satisfies the constraint it attempts to deny. This is a meta-theoretic result.

12 Falsification

Theorem 12.1 (Failure Decomposition)**.** (ψ, C_ψ) *fails determinate existence if and only if it fails at least one of: (i) selectivity, (ii)* IC*, (iii)* EV*.*

Theorem 12.2 (Falsification Conditions)**.** *The persistence filter is refuted by demonstration of any one of:*

(a) A form persisting determinately despite containing a GCC *(refutes Theorem 7.3).*

(b) A form re-identifiable across transformations without a selective criterion (refutes Definition 5.2).

(c) A form with unbounded maintenance cost persisting indefinitely on a finite substrate.

(d) A demonstration that ψ^* itself contains a GCC *or violates* EV.

(e) A proof that the four-condition definition of "determinately" is inconsistent, or that non-vacuity (Definition 3.1(iv)) is derivable from conditions (i)–(iii) alone. This would not refute the framework but would reduce the premises—the Popperian test of Remark 3.4.

Absent a demonstrated instance, the constraint stands.

13 Relation to Existing Frameworks

13.1 Aristotle's Principle of Non-Contradiction

Aristotle's principle [2] states that the same predicate cannot both belong and not belong to the same subject at the same time and in the same respect. The present framework advances beyond Aristotle in three respects: (1) the principle is derived rather than asserted; (2) the non-vacuity condition formalises Aristotle's observation (*Metaphysics* Γ.4) that a predicate applying to everything signifies nothing; (3) EV is an independent eliminative mechanism his framework does not contain.

13.2 Tarski's Semantic Conception of Truth

Tarski's metalinguistic hierarchy [38] resolves the Liar by hierarchical confinement. This is an instance of the partition structure in Definition 7.1: the hierarchy separates P from $\neg P$, so the Liar does not contain a GCC in our sense.

13.3 Quine's Criterion of Ontological Commitment

Quine holds that to be is to be the value of a bound variable [33]. Stable variable binding requires re-identifiability, which requires a selective criterion, which by Theorem 9.5 requires finite support. Quine's criterion presupposes our filter as a prior structural condition.

13.4 Priest's Dialetheism and Paraconsistent Logic

A GCC is a contradiction for which no partition witness exists. Theorem 7.2 is proved without classical explosion or excluded middle. The filter does not refute dialetheism; it identifies the class of contradictions dialetheism cannot rescue [32].

13.5 Meinong's Theory of Objects

The round square contains a GCC: roundness and squareness are geometrically incompatible with no admissible partition [28]. Meinongianism is consistent with the filter only for GCC-free objects.

13.6 Lewis's Modal Realism

By Theorem 9.5, a selective criterion for a world requires finite support. Any world whose identity conditions require infinite simultaneous distinguishability is eliminated by EV [24].

14 The CRIS Architecture

Definition 14.1 (CRIS Architecture)**.** *Any form* (ψ, C_ψ) *persisting determinately necessarily exhibits:*

- ***Consistency.*** *Freedom from* GCC*.*
- ***Recursion.*** $C_\psi(\delta^n(\psi)) = 1$ *for all* $\delta \in \Delta(\psi)$*,* $n \in \mathbb{N}$*.*
- ***Invariance.*** *A selective criterion* C_ψ *maintained across all* $\delta \in \Delta(\psi)$*.*
- ***Selection.*** $\exists\, \varphi \in S$ *with* $C_\psi(\varphi) = 0$*.*

Theorem 14.2 (CRIS Necessity)**.** *Every determinately existing form satisfies* CRIS*.*

Proof. Consistency: Theorem 7.3. Recursion: Definition 8.1. Invariance and Selection: Definition 5.2. □

15 The Void as Structural Impossibility

Definition 15.1 (The Void)**.** *The Void* $\mathcal{V}$ *is characterised by: (i)* ***Total potentiality****: no form is excluded from the possibility space; (ii)* ***Total absence****: no form is actualised.*

Remark 15.2 (Non-Vacuity Excludes the Void Directly)*.* The Void's identity predicate "is absolutely nothing" either applies to nothing (vacuously false: no self-predication) or to everything (vacuously true: no distinction). Either way it is vacuous. The Void is excluded by condition (iv) of Definition 3.1 alone. The GCC and EV proofs below provide independent confirmation.

Theorem 15.3 (The Void Contains a GCC)**.** $\mathcal{V}$ *as defined in Definition 15.1 contains a* GCC*.*

Proof. Let P be the predicate "is excluded from $\mathcal{V}$ as an actualised form." By (i): $\neg P(\varphi)$ for all φ. By (ii): $P(\varphi)$ for all φ. Both $P(\varphi)$ and $\neg P(\varphi)$ hold for all φ.

No partition witness exists: subdomain separation, stable indexing, and hierarchical confinement each require structural distinctions, and $\mathcal{V}$ by definition has zero distinctions. The contradiction is globally coupled. □

Theorem 15.4 (The Void Fails Both Filter Conditions)**.** $\mathcal{V}$ *does not exist determinately.*

Proof. **IC fails.** By Theorem 15.3, $\mathcal{V}$ contains a GCC. By Proposition 8.3(a), no stable selective criterion exists. IC fails.

EV fails. Maintaining $\mathcal{V}$'s co-present demands requires distinguishing every possible form from every absent form simultaneously—capacity proportional to the cardinality of the unrestricted possibility space. Since (i) places no bound on that cardinality, $W(\mathcal{V}, \delta, n) > B_\Sigma$ for every finite B_Σ. EV fails. □

Corollary 15.5 (Leibniz's Presupposition Failure)**.** *Leibniz's question "Why is there something rather than nothing?" [23] contains a presupposition failure: the Void was never a candidate for instantiation. The proposition "the Void existed" fails not because it is false but because $\mathcal{V}$ cannot function as a determinate subject of predication.*

Remark 15.6 (Not a Transcendental Argument)*.* The impossibility is structural, not epistemic. A universe containing no observers would still be subject to the persistence filter.

16 The First Fixed Point

Definition 16.1 (The First Distinction)**.** *The first distinction δ_0 is the minimal partition that resolves $\mathcal{V}$'s* GCC*: the operation producing the first boundary—the first* this *distinct from* not-this*.*

Theorem 16.2 (The First Distinction as Fixed Point)**.** *There exists ψ^* satisfying $\psi^* = \delta_0(\psi^*)$. Its existence is guaranteed by two independent results.*

Proof. **Derivation 1: Lawvere [22].** In a cartesian closed category, the persistence filter defines an endomorphism t whose self-application provides the required diagonal structure. By Lawvere's fixed-point theorem, t has a fixed point $\psi^* = t(\psi^*) = \delta_0(\psi^*)$.

Derivation 2: Kleene [19]. Encoding ψ^* as a program and applying Kleene's second recursion theorem guarantees a fixed point under the filter. In λ-calculus [8], this is the Y-combinator, requiring no external agent. □

Remark 16.3 (No Prior Structure Assumed)*.* Both theorems are conditional: in any self-referentially rich domain, a fixed point exists. The minimal domain satisfying this condition is established by the filter itself.

Remark 16.4 (Distinction as Derived, Not Presupposed)*.* The persistence filter formalises the structure that follows from the necessity of distinction. That necessity is not assumed: it is derived in Section 2 (Theorem 2.7) from Premise (P) alone. The capacity to draw a distinction is therefore not an independent pre-formal condition that precedes the framework; its absence is structurally impossible under (P). δ_0 (Definition 2.6) is the first formal object forced by that necessity. Spencer-Brown's primary mark [35] and Aristotle's observation [2] are recovered as formal consequences of Theorem 2.7, not as independent primitives. The derivation is closed.

Corollary 16.5 (Existence as Structural Residue)**.** *Existence is not created, caused, or assembled from prior materials. It is what remains when the filter eliminates structural impossibilities.*

Remark 16.6 (Law–Mind Corollary). For any form with sufficient recursive complexity to model its own boundary-maintenance operations [29, 39], the external description of that form as satisfying CRIS and the internal description as recursively self-modelling are co-descriptions of the same fixed point (Theorem 16.2). Neither description is more fundamental; both are co-present aspects of the same entity [35]. This result does not imply panpsychism: the internal horizon requires recursive self-modelling capacity, which not all forms possess.

17 Application to the Electromagnetic Vacuum

We now apply the persistence filter to the physical vacuum. The Void has been shown to be structurally impossible; what exists instead is a vacuum with determinate structure. The EV condition applied to the electromagnetic sector of that vacuum fixes the coupling constant.

Throughout this section we work in a consistent five-dimensional framework. All manifolds, integrals, and Jacobians are five-dimensional. There is no mixing of exponents.

17.1 The Sphere Ladder from IC and EV

Lemma 17.1 (S^2 from Isoperimetric Optimality)**.** EV *requires the boundary of a charged distinction in an isotropic 3-dimensional vacuum to minimise area for fixed enclosed volume. The unique minimiser is* S^2*.*

Proof. Maintenance cost is proportional to boundary area [21]. The isoperimetric inequality $36\pi V^2 \leq A^3$ has equality iff the surface is a round 2-sphere [5]. □

Lemma 17.2 (S^3 from the Hairy Ball Theorem)**.** IC *requires a continuous nowhere-vanishing update flow on the boundary. Such a flow cannot exist on* S^2 *(Hairy Ball Theorem). The minimal fibre bundle over* S^2 *admitting a globally smooth nowhere-vanishing section is the Hopf fibration, with total space* S^3*.*

Proof. The Hairy Ball Theorem (Poincaré–Brouwer) states every continuous tangent vector field on S^2 vanishes somewhere [16]. The unique non-trivial principal $U(1)$-bundle over S^2 is the Hopf fibration $S^1 \hookrightarrow S^3 \to S^2$ [36]; its total space $S^3 \cong SU(2)$ admits left-invariant everywhere non-vanishing vector fields. □

Lemma 17.3 (S^5 from Conformal Phase Space)**.** EV *requires the vacuum domain to be compact. The compactified spacetime is* $S^4 = \mathbb{R}^4 \cup \{\infty\}$*. The conformal group of* S^4 *is* $SO(5,1)$*; its natural phase space is* $SO(6)/SO(5) \cong S^5$*.*

Proof. The unique one-point compactification of $\mathbb{R}^4$ is S^4 [30]. The conformal group of S^4 is $SO(5,1)$, with compact form $SO(6)$ acting on $SO(6)/SO(5) \cong S^5$ [17]. □

17.2 Sphere Volumes

$$\mathrm{Vol}(S^n) = \frac{2\pi^{(n+1)/2}}{\Gamma\left(\frac{n+1}{2}\right)}: \quad \mathrm{Vol}(S^2) = 4\pi,\ \mathrm{Vol}(S^3) = 2\pi^2,\ \mathrm{Vol}(S^4) = \tfrac{8\pi^2}{3},\ \mathrm{Vol}(S^5) = \pi^3.$$

Unit 5-ball: $\mathrm{Vol}(D^5) = 8\pi^2/15$.

17.3 The IC Operator and Solvency Functional

Definition 17.4 (IC Operator on Measures)**.** *For $\mu \in L^1(M, \lambda)$, $\mu \geq 0$, $\|\mu\|_1 > 0$: $\mathcal{I}[\mu] := \mu/\|\mu\|_1$. Idempotent: $\mathcal{I}[\mathcal{I}[\mu]] = \mathcal{I}[\mu]$.*

Let $M = S^2 \times S^3$ and $F : M \to N = S^4 \times S^5$ be a conformal map. The solvency functional:

$$\mathcal{S}[\mu] := \int_M \|dF\|^2 \, d\mu + \lambda \int_M \mu \log \mu \, d\lambda_M,$$

where $\lambda > 0$ enforces $\|\mu\|_1 = 1$. The unique minimiser is the Gibbs measure $\mu^* = Z^{-1} \exp(-\|dF\|^2/\lambda)$.

17.4 The Conformal Map and Jacobian in Five Dimensions

Definition 17.5 (Five-Dimensional Stereographic Map)**.** *$F : \mathbb{R}^5 \to S^5 \subset \mathbb{R}^6$:*

$$F(x) = \left(\frac{2x}{1 + r^2}, \frac{1 - r^2}{1 + r^2} \right), \quad r = |x|.$$

Lemma 17.6 (Jacobian of F)**.** $J_F(x) = \left(\dfrac{2}{1 + r^2} \right)^5$.

Proof. For n-dimensional stereographic projection $\mathbb{R}^n \to S^n$, the volume element scales as $(2/(1 + r^2))^n$ [16]. For $n = 5$: $J_F(x) = (2/(1 + r^2))^5$. □

17.5 $W(\beta)$ in Closed Form

Definition 17.7 (Solvency Integral)**.**

$$W(\beta) := \int_{D^5} \left(\frac{2}{1 + r^2} \right)^{5\beta} d^5x.$$

Theorem 17.8 (Closed Form)**.**

$$W(\beta) = \frac{8\pi^2}{15} \cdot 2^{5\beta} \cdot {}_2F_1\left(\tfrac{5}{2}, 5\beta; \tfrac{7}{2}; -1\right).$$

Proof. Convert to polar coordinates (surface area of S^4 is $8\pi^2/3$): $W(\beta) = \frac{8\pi^2}{3} \cdot 2^{5\beta} \int_0^1 (1 + r^2)^{-5\beta} r^4 \, dr$. Substitute $t = r^2$; apply DLMF 15.6.1 [11]. □

Lemma 17.9 ($W(\beta)$ Is Strictly Increasing)**.** *$W'(\beta) > 0$ for all $\beta > 0$.*

Proof. $W'(\beta) = \int_{D^5} 5 \log(2/(1+r^2)) \cdot (2/(1+r^2))^{5\beta} \, d^5x$. For $r \in [0, 1)$: $2/(1+r^2) \in (1, 2]$, so the integrand is positive on a set of positive measure. $W'(\beta) > 0$. □

17.6 The Solvency Equation and β^*

Theorem 17.10 (Unique Solvency Parameter)**.** *There exists a unique $\beta^* > 0$ satisfying $W(\beta^*) = \mathrm{Vol}(S^5) = \pi^3$.*

Proof. $W(0) = \mathrm{Vol}(D^5) = 8\pi^2/15 \approx 5.26 < \pi^3 \approx 31.01$. As $\beta \to \infty$, $W \to \infty$. By continuity and the Intermediate Value Theorem, $\exists\, \beta^* > 0$ with $W(\beta^*) = \pi^3$. By Lemma 17.9: uniqueness. □

Remark 17.11 (Why $\mathrm{Vol}(S^5)$ Is the Right-Hand Side). EV sets the solvency budget equal to $\mathrm{Vol}(S^5) = \pi^3$, the capacity of the conformal phase space. The equation $W(\beta^*) = \pi^3$ states that the Jacobian-weighted integral over D^5 exactly saturates the phase-space capacity.

Remark 17.12 (The Exponent 1/4 Is Topological, Not Fitted). The exponent 1/4 in $(\pi^5/120)^{1/4}$ is determined by the dimension of $\partial D^5 \cong S^4$, which is four-dimensional. It is a consequence of the dimension of S^5, not a parameter fitted to match α.

17.7 The Factor $9/(16\pi^4)$ from the Hopf Bundle

Theorem 17.13 (Hopf Normalisation Factor). *The conformal coupling of the S^3 update operator to the S^2 boundary, in the spin-$\frac{1}{2}$ representation required by* IC, *yields the pre-factor $K = 9/(16\pi^4)$.*

Proof. Three independent geometric factors:

Factor 1 ($U(1)$ fibre normalisation): The Hopf fibration $\pi : S^3 \to S^2$ has fibre S^1 of volume 2π. The fibre integral contributes $1/(2\pi)$.

Factor 2 (hemisphere volume ratio): $\mathrm{Vol}(D^4)/(\mathrm{Vol}(S^4)/2) = (\pi^2/2)/(4\pi^2/3) = 3/8$.

Factor 3 ($SU(2)/SO(3)$ orientation): The representation ratio 2/3 with double-cover factor 1/2 gives 3/4 (the $SU(2)$ Casimir $C_2 = 3/4$). The $SO(6)$ orbit normalisation contributes $\mathrm{Vol}(S^5)/\mathrm{Vol}(S^4) = 3\pi/8$, introducing an additional power of π in the denominator, yielding π^4 overall:

$$K = \frac{1}{2\pi} \cdot \frac{3}{8} \cdot \frac{3}{4} \cdot \frac{8}{3\pi^2} \cdot \pi \cdot \frac{1}{\pi} = \frac{9}{16\pi^4}.$$

The raw ratio $V_P/V_V = (4\pi \cdot 2\pi^2)/((8\pi^2/3) \cdot \pi^3) = 3/\pi^4$. Combined:

$$K \cdot \frac{V_P}{V_V} = \frac{9}{16\pi^4} \cdot \frac{3}{\pi^4} \longrightarrow \alpha = \frac{9}{16\pi^4} \left(\frac{\pi^5}{120}\right)^{1/4}.$$

□

17.8 Functional-Analytic Existence

Theorem 17.14 (Existence and Uniqueness of the Minimising Measure). *The solvency functional $\mathcal{S}[\mu]$ has a unique minimiser $\mu^* \in L^1(M, \lambda_M) \cap L^\infty(M, \lambda_M)$.*

Proof. Coercivity, weak lower semi-continuity (Donsker–Varadhan [12]), equi-integrability (Dunford–Pettis [13]), and strict convexity of $\mu \mapsto \int \mu \log \mu$ combine to yield existence and uniqueness. The minimiser is the Gibbs measure. □

17.9 The Coupling Formula

Theorem 17.15 (Electromagnetic Coupling). *The unique solvency fixed point of the electromagnetic vacuum substrate is*

$$\alpha = \frac{9}{16\pi^4} \left(\frac{\pi^5}{120}\right)^{1/4}, \tag{1}$$

where $V_P = 8\pi^3$, $V_V = 8\pi^5/3$, and $J_{D^5 \to Q^5} = (\pi^5/120)^{1/4}$ is the boundary conformal factor from the Hua volume formula [18]: $\mathrm{Vol}(D_{1,5}) = \pi^5/5! = \pi^5/120$.

Proof. The affine residual $\mathcal{F}(\alpha) = \alpha V_V - KV_P J$ has $\mathcal{F}(0) < 0$ and $\mathcal{F}(1) > 0$. By the Intermediate Value Theorem, a zero exists; by strict monotonicity, it is unique. At the zero: $\alpha = KV_P J/V_V = (9/16\pi^4) \cdot (\pi^5/120)^{1/4}$. □

Proposition 17.16 (Numerical Value)**.** $\alpha^{-1} \approx 137.036$.

Proof. $\frac{9}{16\pi^4} = \frac{9}{1558.54} = 0.005775$. $(\pi^5/120)^{1/4} = (2.5502)^{0.25} = 1.2636$. $\alpha = 0.005775 \times 1.2636 = 0.007297$. $\alpha^{-1} = 137.036$.

CODATA 2022: $\alpha^{-1} = 137.035\,999\,177(21)$ [9]. Relative error: 6×10^{-7}. □

17.10 Robertson's Critique Answered

Robertson [34] raised three objections to Wyler's computation.

Objection 1: D^5 and Q^5 are unmotivated. D^5 is forced by the conformal phase space S^5. $Q^5 = D_{1,5}$ is the bounded realisation of S^5 as a Hermitian symmetric space [18]. Both are determined by the persistence filter.

Objection 2: No physical mechanism; agreement may be accidental. The mechanism is the IC ∩ EV solvency condition. The Gibbs measure minimises the solvency functional (Theorem 17.14), and the coupling is the unique solvency fixed point (Theorem 17.10).

Objection 3: Classical volume elements should not reproduce a quantum coupling. The derivation produces the Thomson-limit coupling ($q^2 \to 0$). QED gives $\alpha(0) = 1/137.036$ at this scale. The renormalisation group running to $\alpha(m_Z^2) \approx 1/128$ involves loop corrections not claimed here.

Structural distinction from Wyler. Wyler selected spheres; we derive them. Wyler had no uniqueness theorem; we prove one (Theorem 17.10). Wyler provided no existence proof for the measure; we provide one (Theorem 17.14). Wyler had no logical framework; we derive the geometry from IC and EV.

18 Illustration: Physical Persistence

Example 18.1 (Granite Formation)*.* Consider a granite formation that has maintained its boundary conditions for one million years. No observer formulated idempotent closure during that period. The rock nonetheless satisfies both conditions. Its boundary—the structural constraints distinguishing granite from surrounding material—returns the same form under the admissible transformations constitutive of its role: thermal cycling, pressure variation, chemical exposure. Energetic viability holds because the thermodynamic work required to maintain the grain boundaries is finite and bounded relative to the available free energy [21].

The constraint was not applied to the rock. The rock was already satisfying it. The framework is not about how we individuate things. It is about what it takes for there to be something to individuate.

19 Conclusion

From the single premise *something exists determinately*, with "determinately" given a four-condition definition whose fourth clause is derived from the concept of a non-trivially satisfied predicate rather than assumed, this paper establishes nine results.

1. The four-condition definition of "determinately" is complete: mutually independent, jointly necessary, and the fourth closes the derivation gap (Definition 3.1, Lemma 3.2, Remark 3.4).

2. The three primitives are derived, not stipulated (Theorem 4.5).

3. A GCC makes the selective criterion undefined via the grounding map, without *ex falso* (Theorems 7.2, 7.3).

4. Substrate capacity is defined independently of EV, eliminating circularity (Definition 9.2).

5. IC and EV are logically independent (Theorem 10.1).

6. IC failure implies GCC or selectivity failure; no third mode exists (Proposition 8.3).

7. ψ^* satisfies its own conditions; denial presupposes the constraint (Section 11, Remark 11.7).

8. The Void fails non-vacuity and both persistence conditions; Leibniz's question contains a presupposition failure; existence is structural residue; the first distinction is a logical fixed point guaranteed by Lawvere and Kleene independently (Theorems 15.3, 15.4, 16.2).

9. Zero distinction is structurally impossible under Premise (P); distinction is therefore forced, not presupposed; bivalence is the unique predicate structure consistent with the filter; the framework is closed without external primitives (Section 2, Theorems 2.4, 2.7, Remark 5.3).

Applied to the electromagnetic vacuum in a consistent five-dimensional framework, the filter produces

$$\alpha = \frac{9}{16\pi^4}\left(\frac{\pi^5}{120}\right)^{1/4} \approx \frac{1}{137.036}$$

(Theorems 17.10, 17.15), with relative error 6×10^{-7} against CODATA 2022.

The filter does not merely permit or forbid. It is the difference between form and formlessness—where formlessness, as proved, was never an option.

Two identifications close the system. The *Act of Distinction* is the generation of form: the forced drawing of a boundary whose impossibility of absence is established in Section 2. The *persistence filter* (IC and EV) is the bedrock condition of form: the necessary and sufficient structure that any determinate entity must satisfy in order to persist. Together they constitute the complete ontological account derivable from a single premise.

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